Escape the Network Learning Resources

Use these resources to learn the theory before building your route planner. Work through the videos first, then practise with the interactive visualisations. Each underlined resource title is a clickable link.

1 Stacks and queues

Craig n Dave — Stacks and queues

Start with FIFO and LIFO. Understand why a queue supports breadth first search and a stack supports depth first search.

2 Graph traversal with BFS and DFS

Craig n Dave — Graph traversal with BFS and DFS

Watch worked explanations of both searches and their comparison. Pause to predict which node will be processed next.

3 Dijkstra shortest path algorithm

Craig n Dave — Dijkstra shortest path algorithm

Learn how tentative distances and predecessors are updated to find the route with the lowest total cost.

4 Interactive BFS and DFS

VisuAlgo — Interactive BFS and DFS

Use an example graph and step through BFS and DFS. Compare the exploration order and practise predicting the next step.

5 Interactive Dijkstra

VisuAlgo — Interactive Dijkstra

Select Dijkstra from the available algorithms. Follow changes to tentative distances on a weighted graph.

Suggested way to study

Watch an explanation, predict the next step, check the animation, then trace a new graph independently. Record the frontier, visited or settled nodes, predecessors and tentative distances where relevant.

The videos are labelled OCR or AQA, but the graph-search concepts support this project. VisuAlgo includes additional algorithms; focus on BFS, DFS and Dijkstra for now.